NOVA IMS

INFORMATION MANAGEMENT SCHOOL

Technical Proposal Guidelines

Due Sunday, 6 December 2026, at 23:59

NOVA IMS

Bachelor's Degree in Data Science

Capstone Project

2026-2027

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Due **Sunday, 6 December 2026, at 23:59**.

The team's representative submits it on Moodle, once for the whole team, as a PDF named `TeamName_TechnicalProposal.pdf`.

The technical proposal is the document a technical lead would ask for before the last month of a project: what you are building, how it is built, what works today and what will be done by Sunday 3 January. It is assessed together with the Week 3 proposal, and the final submission is compared with it.

There is no page limit, but padding has a cost: text that adds nothing makes the document less clear. Cover each section fully.

Use the sections below, in this order.

1. Cover and contents

Team name, every member's name and student number, the date, and a table of contents.

2. Summary

One page at most: the problem, the users, how the application solves it, and what works today.

3. Since the Week 3 proposal

What you kept, what you changed and why. Answer the feedback you received on the proposal, point by point.

4. The problem and the solution

The problem, who has it, what exists for it today and what yours does differently. What a user can do with the application, and what the AI does in it that ordinary code could not.

5. Architecture

The core of the document. A diagram showing the layers, every component in them, and the path of one request through them. Then, for each required component of the Project Guidelines, how your application meets it:

- the model calls, and where their answers come back as structured output
- the agent: what the conversation can change, the tools it chooses from, and one request that chains several calls
- the tools: each one's purpose; which ones write, and which ask the user first
- the documents the application reads, and what it does with their content
- what each layer contains, and where configuration and prompts live
- tracing: what the trace of one request contains, whether prompts and responses are recorded, and for how long traces are kept

Close with the external services the application depends on, the personal data it handles, if any, and where users are told they are talking to an AI system.

Diagrams are labelled and captioned.

6. Technical choices

Every major choice and why it serves your project, with its trade-offs: provider and models, interface, storage, tracing and deployment, whether taught in the course or researched by you.

7. State and scope

Three lists: what works today, with known issues; what will be done by 3 January, in priority order; what you considered and left out, with the reason.

8. Plan to 3 January

The steps from here to 3 January, each with what it delivers and its date. How you plan is your decision; the plan is judged on whether it is realistic.

9. Team

Each member's role and what they own. How you organise the work is also your decision; looking at how software teams do it is worth the time.

10. Risks and open decisions

What has gone wrong so far and what you did about it. What could go wrong before 3 January and what you will do. Decisions still open, with their options.

What not to include

Code, which lives in the repository. Explanations of concepts taught in the course. Marketing language. Placeholder sections.

How it is assessed

- **Clarity.** Structured, easy to follow, written for a technical reader.
- **Technical depth.** The architecture is understood and the choices are justified; the diagrams match the design.
- **Completeness.** Every section is present, at the right level of detail.
- **Feasibility.** The plan is realistic for the time and the team, and the risks are real ones.
- **Professionalism.** Formatting, grammar and tone.

At the final submission, what was delivered is compared with sections 7 and 8.